

SYNERGISTIC ACTIVITIES

- **Research Conference Organizing:** The PI has organized many national research conferences and workshops. Most recently the PI was a co-organizer for the Fall 2024, *Algebraic Geometry Northeastern Series (AGNES)*, a weekend conference with 100 participants held at Dartmouth College aimed at introducing early career researchers (e.g. graduate students and postdocs) to current research trends in algebraic geometry and related fields.
- **Developed Open-Source Software:** The PI has a long commitment to developing open-source software to promote and support research in algebraic geometry and commutative algebra. For example, the PI developed the *VirtualResolutions* package for the computer algebra system *Macaulay2*. This package, which includes core functionality for multigraded homological algebra on toric varieties, has been proven beneficial to many research projects and is frequently cited.
- **Invited Research Talks:** The PI has given many invited research talks including giving a plenary talk *Homological Algebra Beyond Projective Space* at $\text{Spec}(\overline{\mathbb{Q}}(2\pi i))$ a research conference held at the Fields Institute.
- **Preparing Future Researchers** The PI has organized and lead numerous initiatives and conferences aimed at helping graduate students and early career researchers establish themselves within the field, find research mentors, and a community of collaborators. Of note is the *Gems Of...* series of weekend and week-long workshops that the PI organized, with NSF support, for young researchers in commutative algebra and combinatorics. This series has reached over 150+ participants over 3 years.
- **Growing the Mathematics Community** The PI served as the inaugural president for *Spectra*, aimed at widening the mathematical community. This included growing the organization's budget by \$20,000 and its membership to over 500 people, and launching the organization's annual invited lecture series at the Joint Mathematics Meeting.